

Xiaosheng Huang

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Education

PhD, Physics: University of California, Berkeley	December, 2004
MA, Physics, Kent State University	May, 1995
BS, Physics (<i>Summa Cum Laude</i>), Kent State University	May, 1995

Professional Appointments

John Nobili, SJ Professor of Physics: Santa Clara University	2026 –
Professor: Department of Physics & Astronomy, University of San Francisco	2024 – 2026
Associate Professor: Department of Physics & Astronomy, University of San Francisco	2018 – 2024
Assistant Professor: Department of Physics & Astronomy, University of San Francisco	2012 – 2018
Lecturer: Physics Department, University of California, Berkeley	2007 – 2012
Postdoctoral Fellow: UC Berkeley/Lawrence Berkeley National Laboratory	2005 – 2012
Graduate Student Researcher: Physics Department, UC Berkeley	2000 – 2004
Graduate Student Researcher: Astronomy Department, UC Berkeley	1996 – 1999
Graduate Student Instructor: Physics Department, UC Berkeley	1995, 1999

Press Releases & Publicity

American Astronomical Society Nova Highlight	2026
USF News Article on Two Hubble Space Telescope Programs	2026
NPR Interview	2025
Lawrence Berkeley National Laboratory Press Release	2024
USF News Article on Roman Space Telescope Program	2023
Lawrence Berkeley National Laboratory Computing Sciences Highlight	2023
European Southern Observatory Press Release	2023
American Astronomical Society Nova Highlight	2023
Press Conference, American Astronomical Society Annual Meeting	2021
American Astronomical Society Nova Highlight	2021
National Science Foundation's NOIRLab Press Release	2021
Lawrence Berkeley National Laboratory Press Release	2021
Lawrence Berkeley National Laboratory Press Release	2020
USF Magazine Article on Faith	2020
USF News Article on Hubble Space Telescope Program	2019

Awards & Grants

Distinguished Research Award , University of San Francisco	2026
Hubble Space Telescope Program GO-18085 , with NASA Grant (\$128K)	2025
Hubble Space Telescope Program AR-18151 , with NASA Grant (\$29K)	2025
Dean's Scholar Award , College of Arts & Sciences, USF (\$10K)	2024
NASA Roman Space Telescope Mission Research and Support Program (\$64K)	2023
Outstanding Mentors Award , Lawrence Berkeley National Laboratory	2021
Hubble Space Telescope Program GO-15867 , with NASA Grant (\$189K)	2019
Visiting Faculty Program , Department of Energy (with summer stipends)	2017, 2018, 2019, 2025
Faculty Development Fund, USF (multiple awards each year)	2012 –
Graduate Student Travel Grant, University of California, Berkeley	2004
University Fellowship, University of California, Berkeley	1995

Research

Research Highlights

Among the biggest questions in cosmology are the nature of dark energy and dark matter and the expansion rate of the universe. To address these questions, I lead the **multi-institutional DESI Strong Lens Foundry** within the [Dark Energy Spectroscopic Experiment \(DESI\)](#) and collaborate closely with the *Supernova Cosmology Project* led by Prof. Saul Perlmutter, both based at Lawrence Berkeley National Laboratory (LBNL). Using [DESI imaging data](#), my team has discovered over 5000 strong lens candidates, featured in multiple press releases and a national press conference. We also found 1346 lens candidates in the [UNIONS](#) imaging data. In DESI spectroscopy, using “classical” methods, we found an additional 6000 candidates, reaching a grand total of 12,000. We developed a GPU-accelerated Bayesian lens modeling framework that underpins our current and forthcoming Hubble Space Telescope (HST) and JWST programs. I am PI on three HST programs with NASA support (\$189K in 2019; \$157K in 2025) and supervise two additional programs for which I serve as Co-I. I am also Co-PI/Co-I on major observing programs at the Gemini, Keck, and Very Large Telescopes, as well as on space-based programs with the Nancy Grace Roman Space Telescope (\$64K NASA grant) and JWST. We were recently selected by the competitive [NERSC](#) Science Acceleration Program ([NESAP](#)), with early access to the next-generation supercomputer [Doudna](#), with 12,000 GPU hours awarded and NERSC engineers dedicated to helping adapt and scale [GIGA-Lens](#) to the new architecture. Finally, we are preparing a targeted search for lensed supernovae to measure H_0 , with a paper demonstrating accurate and precise modeling of lensed point sources using [GIGA-Lens](#) ([arXiv:2601.18787](#)) now submitted.

Selected publications and lens candidate catalogues are available at our [project website](#).

Principal Investigator, Submitted Proposal

- PI, JWST, 2025 (Cycle 5): The Cosmic Carousel: A JWST Experiment for Dark Energy and Dark Matter with a Unique Cluster Strong Lens

Principal Investigator, Successful Telescope Proposals

- PI, HST, 2025 (Cycle 33): [Confirming Spectroscopic Strong Lenses Found in New Discovery Windows from DESI DR1](#)
- PI, HST, 2025 (Cycle 33): [Finding Low Halo-Mass Strong Lenses in the HST Archive](#)
- Co-PI, [Keck Observatory](#), 2022–2025: Redshifts and Modeling for Low-Mass Halo Detection
- Co-PI, [DESI](#) Secondary Target Program, 2021: Spectroscopic Confirmation of Lens Candidates
- PI, HST, 2019 (Cycle 27): [Confirming Strong Galaxy Lenses in DESI Legacy Surveys](#)

Co-Investigator, Successful Telescope Proposals

- HST:
 - 2024, [Confirming Spectroscopic Strong Lens Candidates from the DESI One-Percent Survey](#) (PI: A. Bolton)
 - 2024, [Automated Pipeline for Modeling Strong Lenses Observed by HST](#) (PI: C. Storfer)
 - 2020, [LensWatch: Time-Delay Measurement of a Multiply-Imaged Supernova](#) (PI: J. Pierel)
 - 2016–2017, [SUBaru Supernovae with Hubble Infrared \(SUSHI\)](#) (PI: N. Suzuki)
 - 2014–2015, [See Change: Testing Time-Varying Dark Energy with \$z > 1\$ Supernovae](#) (PI: S. Perlmutter)
- [Alma APEX](#): 2024, ASADOS (PI: K. Harrington)

- Gemini: 2024, Gravitational Lenses in UNIONS and Euclid (GLUE): Spectroscopic Confirmation of Strongly Lensed Galaxies (PI: C. Storfer)
- Nancy Grace Roman Space Telescope: 2023 ([NASA ROSES-D.14](#)), Enabling Lensed SN Cosmology (PI: J. Pierel)
- Joint HST + JWST Program (2023): [Pioneering Precision: Advancing Cosmology with the First Statistical Sample of Lensed Supernovae](#)
- JWST (2022, Cycle 1): [Imaging and Spectroscopy of Three Highly Magnified Images of a Supernova at \$z = 1.5\$](#) (PI: P. Kelly)
- VLT / MUSE: DESJ0603: A uniquely spherical, multi-source lensing cluster perfect for cosmology (PI: T. Barone, 2024)
- VLT / MUSE: Characterization of galaxy-galaxy gravitational lens systems with MUSE (PIs: A. Cikota, F. Bian 2021 - 2023)
- VLT / Xshooter: Strong lensing and mass assembly (2021–2022, PI E. Jullo)
- [NSF NOIRLab DECam / Blanco](#): Imaging and transient surveys (2019–2023; PIs D. Schlegel, A. Palmese, I. Andreoni)
- Keck Observatory: Multiple programs, 2010–2018

Principal Investigator, Successful Supercomputer Proposals

- PI, 2026–2027, [NESAP](#) selection
- Co-PI, multiple proposals 2020–2026, NERSC “regular” allocations ([ERCAP](#))

Teaching

Teaching Highlights

I developed and continue to update the ML/AI course series PHYS 301–303 and founded two minor programs: *Engineering Physics* (2016–) and *Applied AI* (launching 2025). Courses have been adapted/cross-listed at the master’s level across Computer Science and Economics.

Courses, University of San Francisco (2012–)

USF	PHYS 120/121 (Intro. to Astronomy); 301–303 (AI series); 343 (Cosmology); 350 (Colloquium); 371 (Mathematical Methods for Physicists); ENGR 302 (Computation).
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Courses, University of California, Berkeley (2007–2012, Lecturer)

Berkeley	Lower Div: Honors 7B (Gen. Phys. II), 7C (Gen. Phys. III), 8A (Pre-med Gen. Phys.); Upper Div: 112 (Stat Mech), 137 (Quantum Mechanics)
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Development of New Courses

- **PHYS 301: Introduction to Scientific Computation** (2013; major revision 2015). Python-based modernization; added lab; expanded to Monte Carlo, DFT, PCA, SVM; M.S. cross-listing in CS (2022).
- **PHYS 302: Scientific Computation & Machine Learning** (2016). Added decorators, regression, ANNs, CNN, ODE systems, symbolic computation; M.S. cross-listings in CS/Econ (2020, 2023; RL/ResNets).

- **PHYS 303: Deep & Bayesian Learning** (2023). Deep architectures (e.g., Transformer) and Bayesian inference (HMC, VI, latent variables, high-dimensional optimization); M.S. adaptation for CS (2025).
- **ENGR 302: Scientific Computation for Engineers** (2022). Two-unit engineering applications course.
- **PHYS 343: Astrophysics (Cosmology)** (2013). New cosmology course under existing number.

Service

Service Highlights

Mentored **50+** undergraduate researchers at USF, UC Berkeley and other universities, leading to peer-reviewed publications and national presentations; received LBNL’s [Outstanding Mentors Award](#) (2021). Lead the multi-institutional *DESI Strong Lens Foundry* collaboration.

Service to University/College

- **Curriculum:** Engineering curriculum development (2016–); joint appointment (2020–2024).
- **Committees:** Dean’s Award Review (2025); Distinguished Research (member 2014; Chair 2015–2016); Honorary Degree (2014–2018); Learning Technologies (2015–2018); Learning Analytics (2017–2018).
- **Outreach:** Destination USF Mock Class (2016).

Service to Department/Students

- **Mentoring:** Research Supervision since 2013; multiple ApJ/ApJS publications with students (undergraduate and graduate) co-authors/lead authors; CS M.S. capstone mentor (2018, 2019, 2022).
- **Student Presentations:** AAS (incl. 2020); USF CARD (2014, 2015, 2018, 2019, 2023, 2024); USF events: Dean’s Circle (2022), President’s Inauguration (2014), Innovation Symposium (2013).
- **Fellowships:** Guided successful USF Summer Research Fellowship proposal (2015).
- **Programs:** Engineering Physics Minor (2017–; 30+ recipients); Applied AI Minor (launching 2025).
- **Hiring:** Faculty search committees (tenure lines): 2016, 2018.

Professional Activities & Service

- Lead, *DESI Strong Lens Foundry* (2019–).
- LBNL student mentorship (2013–; DOE [SULI](#)); LBNL *Outstanding Mentors Award* (2021).
- PhD Thesis Committee, IfA (U. Hawai’i, 2023–); Senior Thesis Committee, Yale (2024–2025).
- HST proposal reviewer (MidCycle 27; Cycle 29).
- Journal referee: *Nature Astronomy*, *Space Science Review*, *ApJS*, *AJ*, *MNRAS*.
- Invited talks: UC Berkeley Astronomy Colloquium (2025); Santa Clara Univ. Physics Colloquium (2025); UC Davis Cosmology Seminar (2025); University of Rochester Colloquium (2024); USF Colloquium (2023); LBNL Machine Learning Seminar (2022); KIPAC/Stanford Cosmology Seminar (2022); NERSC Data Seminar (2022); LBNL Physics Division Seminar (2021); Florida State University Astrophysics Seminar (2020); USF Colloquium (2017); San Jose State Univ. Colloquium (2016); Wonderfest Public Lecture (2016); Sonoma State Univ. “What Physicists Do” Series (2015).

- Conferences Presentations: Science with Strong Lensing, ALMA and Next-Generation Radio Interferometry (NRAO, Charlottesville, 2025); Strong Gravitational Lensing Science Conference (CfA/Harvard-Smithsonian, 2025); Strong Lensing with LSST (Oxford, 2024); DESI Collaboration Meeting Plenary (2024); American Astronomical Society Annual Meetings (2014–2021); Nearby Supernova Factory Collaboration Meeting (Clermont-Ferrand, 2017).

*Peer-reviewed Publications*¹

1. C. Storfer, K.C. Wong, A. Acebron, C. Grillo, W.B. Hoogendam, **X. Huang**, et al. 2026, submitted to *ApJ*, [arXiv:2604.02418](#)
Supernova 2025wny: High-angular Resolution Keck/NIRC2 Observations and Preliminary Lens Modeling
2. *S. Agarwal[†], **X. Huang**, et al. 2026, *ApJ*, 1001, 162
DESI Strong Lens Foundry III: Keck Spectroscopy for Strong Lenses Discovered Using Residual Neural Networks
3. F. Urcelay, **X. Huang**, et al. 2026, submitted to *ApJ*, [arXiv:2602.16077](#)
The Carousel Lens II: Cosmological Constraints with GIGA-Lens
4. J. O'Donnell, et al. 2026, submitted to *ApJ*, [arXiv:2601.14422](#)
The Carousel Lens I: A Spectroscopic Survey of the Carousel Lens Field
5. **X. Huang**, *S. Baltasar[†], *N. Ratier-Werbin[†], et al., 2026, *ApJ*, 998, 69
DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens
6. *S. Baltasar[†], *N. Ratier-Werbin[†], **X. Huang**, et al. 2026, submitted to *ApJ*, [arXiv:2601.18787](#)
A Novel Lensed Point Source Modeling Pipeline using GIGA-Lens with Application to SN Zwicky and SN iPTF16geu
7. **X. Huang**, *D. Alvarez[†], *M. Ubeda[†], et al. 2025, submitted to *ApJ*, [arXiv:2512.07823](#)
DESI Strong Lens Foundry V: An HST Sample of Strong Lenses Modeled with GIGA-Lens
8. *J. Karp[†], D.J. Schlegel, **X. Huang**, et al. 2026, *ApJ*, 999, 198
The DESI Single Fiber Lens Search I: Four Thousand Spectroscopically Selected Galaxy-Galaxy Gravitational Lens Candidates
9. *E. Lin[†], I. Bertolla, A. Cikota, **X. Huang**, et al. 2025, submitted to *ApJS*, [arXiv:2509.18078](#)
DESI Strong Lens Foundry IV: Spectroscopic Confirmation of DESI Lens Candidates with VLT/MUSE
10. **X. Huang**, *Jose Inchausti, C. Storfer, *S. Tabares-Tarquinio, et al. 2025, submitted to *ApJ*, [arXiv:2509.18089](#)
DESI Strong Lens Foundry II: DESI Spectroscopy for Strong Lens Candidates
11. *Y. Hsu[†], **X. Huang**, et al. 2026, *ApJS*, 282, 41
A New Way to Discover Strong Gravitational Lenses: Pairwise Spectroscopic Search from DESI DR1
12. *Jose Inchausti, C. Storfer, **X. Huang**, et al. 2025, submitted to *ApJ*, [arXiv:2508.20087](#)
Strong Lens Discoveries in DESI Legacy Imaging Surveys DR10 with Two Deep Learning Architectures
13. C. Storfer, E.A. Magnier, **X. Huang**, et al. 2026, *ApJ*, 999, 1
Gravitational Lenses in UNIONS and Euclid (GLUE) I: A Search for Strong Gravitational Lenses in UNIONS with Subaru, CFHT, and Pan-STARRS Data

¹* = Papers led by a student who was an undergraduate at least at the start of the project. An underline indicates undergraduate co-authors from USF, and a [†], those from other institutions.

14. T. Barone, et al. 2025, AJ accepted, [arXiv:2503.08041](#)
The AGEL Survey Data Release 2: A Gravitational Lens Sample for Galaxy Evolution and Cosmology
15. *E. Silver[†], R. Wang[†], **X. Huang**, A. Bolton, et al., 2025, ApJ, 994, 117
ML-Driven Strong Lens Discoveries: Down to $\theta_E \sim 0.03''$ and $M_{\text{halo}} < 10^{11} M_{\odot}$
16. D. Rubin et al. 2025, ApJ, 986 231
Union through UNITY: Cosmology with 2000 SNe Using a Unified Bayesian Framework
17. B. Wedig et al. 2025, ApJ, 986, 42
The Roman View of Strong Gravitational Lenses
18. E. Khalouei et al. 2025, A&A, 698, 266
Detection of unresolved strongly lensed supernovae with the 7-Dimensional Telescope
19. F. Urcelay et al. 2025, A&A, 694, A35
A compact group lens modeled with GIGA-Lens: Enhanced inference for complex systems
20. J.D.R. Pierel et al. 2024a, ApJ, 967, 50
JWST Photometric Time-delay and Magnification Measurements for the Triply Imaged Type Ia “SN H0pe” at $z = 1.78$
21. J.D.R. Pierel et al. 2024b, ApJL 967 L37
Lensed Type Ia Supernova “Encore” at $z = 2$: The First Instance of Two Multiply Imaged Supernovae in the Same Host Galaxy
22. W. Sheu, A. Cikota, **X. Huang** et al., 2024, ApJ, 973, 3
The Carousel Lens: A Well-Modeled Strong Lens with Multiple Lensed Sources
23. *W. Sheu[†], **Xiaosheng Huang**, et al., 2024, ApJ, 973, 24
A Targeted Search for Variable Gravitationally Lensed Quasars
24. *C. Storfer, **X. Huang**, et al., 2024, ApJS, 274, 16
New Strong Gravitational Lenses from the DESI Legacy Imaging Surveys Data Release 9
25. *C. Dawes[†], C. Storfer, **X. Huang**, G. Aldering, A. Dey, D. J. Schlegel, 2023, ApJS, 269, 61
Finding Multiply-Lensed and Binary Quasars in the DESI Legacy Imaging Surveys
26. A. Cikota, I. Bertolla, **X. Huang**, et al., 2023, ApJL, 953, L5
A New Einstein Cross Spectroscopically Confirmed with VLT/MUSE and Modeled with GIGA-Lens
27. A. Filipp, Y. Shu, R. Pakmor, S. H. Suyu, and **X. Huang**, 2023, A&A, 677, A113
Simulation-guided Galaxy Evolution Inference: A Case Study with Strong Lensing Galaxies
28. D. Langeroodi, et al., 2023, ApJ, 957, 39
Evolution of the Mass-Metallicity Relation from Redshift $z \approx 8$ to the Local Universe
29. *W. Sheu[†], **X. Huang**, A. Cikota, N. Suzuki, D.J. Schlegel, C. Storfer, ApJ, 952, 10, 2023
Retrospective Search for Strongly Lensed Supernovae in the DESI Legacy Imaging Surveys
30. J.D.R. Pierel, N. Arendse, S. Ertl, **X. Huang**, L. A. Moustakas, S. Schuldt, A. J. Shajib, Y. Shu, S. Birrer, M. Bronikowski, J. Hjorth, S. H. Suyu, et al., 2023, ApJ, 948, 115
LensWatch: I. Resolved HST Observations and Constraints on the Strongly-Lensed Type Ia Supernova 2022qmx (“SN Zwicky”)
31. *A. Gu[†], **X. Huang**, et al., ApJ, 935, 17, 2022
GIGA-Lens: Fast Bayesian Inference for Strong Gravitational Lens Modeling

32. B. Hayden, et al., ApJ, 912, 87, 2021
The HST See Change Program: I. Survey Design, Pipeline, and Supernova Discoveries
33. **X. Huang**, C. Storfer, A. Gu, V. Ravi, A. Pilon, et al., ApJ, 909, 27, 2021
Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys
34. **X. Huang**, M. Domingo, A. Pilon, V. Ravi, C. Storfer, D.J. Schlegel, et al., ApJ, 894, 78, 2020
Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey
35. J. Kim, M.J. Jee, S. Perlmutter, B. Hayden, D. Rubin, **X. Huang**, G. Aldering, J. Ko, ApJ, 887, 76, 2019
Precise Mass Determination of SPT-CL J2106-5844, the Most Massive Cluster at $z > 1$
36. R. Rubin, B. Hayden, X. Huang, ..., Z. Raha, et al., ApJ, 866, 65, 2018
The Discovery of a Gravitationally Lensed Supernova Ia at Redshift 2.22
37. **X. Huang**, G. Aldering, M. Biederman, and B. Herger, ApJ, 850, 84, 2017
On the Time Variation of Dust Extinction and Gas Absorption for Type Ia Supernovae Observed through a Nonuniform Interstellar Medium
38. **X. Huang**, Z. Raha, et al., ApJ, 836, 157, 2017
The Extinction Properties of and Distance to the Highly Reddened Type Ia Supernova SN 2012cu
39. J. Nordin, et al., MNRAS, 440, 2742, 2014
Lensed Type Ia supernovae as probes of cluster mass models
40. D. Rubin, et al., ApJ, 763, 35, 2013
Precision Measurement of The Most Distant Spectroscopically Confirmed Supernova Ia with the Hubble Space Telescope
41. N. Suzuki, et al., ApJ, 746, 85, 2012
The Hubble Space Telescope Cluster Supernova Survey: V. Improving the Dark Energy Constraints Above $z > 1$ and Building an Early-Type-Hosted Supernova Sample
42. K. Barbary, et al., ApJ, 745, 31, 2012
The Hubble Space Telescope Cluster Supernova Survey. VI. The Volumetric Type Ia Supernova Rate
43. K. Barbary, et al., ApJ, 745, 32, 2012
The Hubble Space Telescope Cluster Supernova Survey: II. The Type Ia Supernova Rate in High-Redshift Galaxy Clusters
44. M.J. Jee, et al., ApJ, 737, 59, 2011
Scaling Relations and Overabundance of Massive Clusters at $z \gtrsim 1$ from Weak-lensing Studies with the Hubble Space Telescope
45. **X. Huang**, et al., ApJL, 707, L12, 2009
Hubble Space Telescope Discovery of a $z = 3.9$ Multiply Imaged Galaxy Behind the Complex Cluster Lens Warps J1415.1+36 at $z = 1.026$
46. K.S. Dawson, et al., AJ, 138, 1271, 2009
An Intensive HST Survey for $z > 1$ Type Ia Supernova by Targeting Galaxy Clusters
47. K. Barbary, K., et al., ApJ, 690, 1358, 2009
Discovery of an Unusual Optical Transient with the Hubble Space Telescope
48. **X. Huang**, W. Mickelson, B.C. Regan, Steve Kim[†], A. Zettl, 2006, Solid State Communications, 140, 163
Pressure Dependence of T_c and H_{c2} of a Dirty Two-gap Superconductor, Carbon-doped MgB_2

- 49. **X. Huang**, W. Mickelson, B.C. Regan, and A. Zettl, 2005, Solid State Communications, 136, 278
Enhancement of the Upper Critical Field of MgB₂ by Carbon-doping
- 50. W. Han, J. Cumings, **X. Huang**, et al., Chemical Physics Letters, 346, 368, 2001
Synthesis of Aligned BxCyNz Nanotubes by a Substitutionreaction Route
- 51. T. Broadhurst, **X. Huang**, B. Frye, and R. Ellis, ApJL, 534, L15, 2000
A Spectroscopic Redshift for the Cl 0024+16 Multiple Arc System: Implications for the Central Mass Distribution

Publications in Prep

- 52. C. Storfer, M. Tamargo[†], **X. Huang**, et al. 2025
DESI Strong Lens Foundry VI: Keck Spectroscopy for 15 Strong Lenses Discovered Using Residual Neural Network
- 53. *N. Ratier-Werbin[†], *H. Lu[†], *L. Upton[†], **X. Huang**, et al. 2025, in prep.
Lens Modeling with Multiple GPU Nodes
- 54. C. Storfer, **X. Huang**, et al. 2026, in prep.
DESI Strong Lens Foundry VII: A Catalog of 1000 Spectroscopically Confirmed Strong Gravitational Lenses in DESI Year 3 Data Release

Other Publications

- Pierel, J. et al., Transient Name Server AstroNote 2022-196 Multiple Images of SN 2022qmx (“SN Zwicky”) Resolved in HST Observations
- Boone, K. et al., 2016ATel. 9125 SCP16L01: discovery of an unusual transient in MOO-J1142